Machine Learning Assignment 2

QSAR Biodegradation Classifier

BITS WILP M.Tech AIML

1. Problem statement

Given molecular descriptors for an organic compound, predict whether it is **readily biodegradable (RB)** or **not readily biodegradable (NRB)**. The application accepts a labelled CSV test set, loads one of the pre-trained models, and reports its performance without retraining on uploaded data.

2. Dataset description

This project uses the **QSAR biodegradation** dataset from the UCI Machine Learning Repository .

- 1,055 compounds and 41 numerical molecular-descriptor features.
- Binary target: RB (readily biodegradable) and NRB (not readily biodegradable).
- The supplied data is stored as `data/qsar_biodegradation.csv`.
- An 80/20 stratified holdout split uses `random_state=2026`. The labelled holdout used for the results is `data/test_data.csv`.

The descriptor names are neutral (`descriptor_01` through `descriptor_41`) because the original UCI file provides descriptor values without a header row. The original source remains the authoritative definition of each descriptor.

3. GitHub repository link

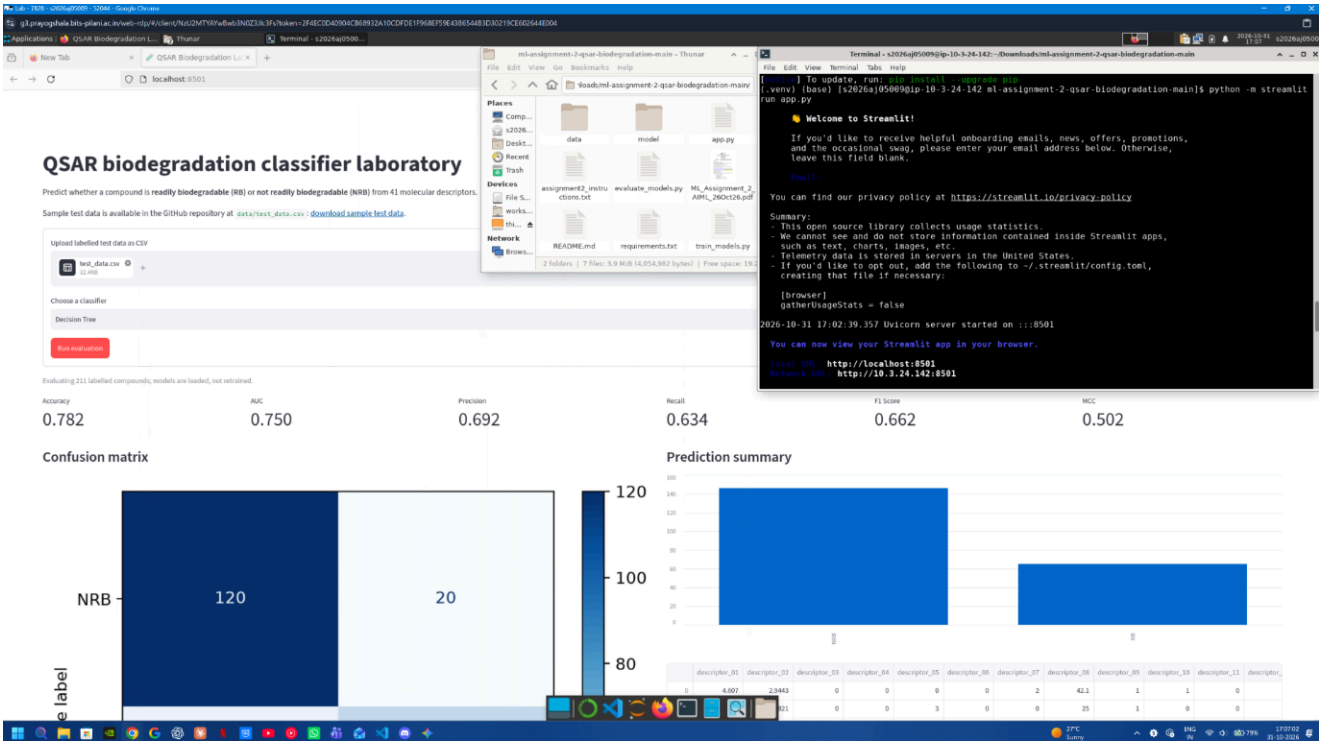
> **Repository:** https://github.com/saurabhtayal/ml-assignment-2-qsar-biodegradation

4. Live Streamlit app link

> **Live application:** https://ml-assignment-2-qsar-biodegradation-onsuyyappeeggmgshgvpift.streamlit.app/

5. BITS Virtual Lab execution screenshot

The screenshot below shows the assignment execution on BITS Virtual Lab.



6. Models used and evaluation

All models were trained on the same training partition and evaluated on the same untouched 211-row test partition. Standard scaling is included in the Logistic Regression, KNN, and SVM pipelines. The Decision Tree and Gaussian Naive Bayes models use the original numerical values.

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.8673	0.8925	0.8308	0.7606	0.7941	0.6980
Decision Tree	0.7820	0.7501	0.6923	0.6338	0.6618	0.5025
KNN	0.8152	0.8570	0.7162	0.7465	0.7310	0.5906
Naive Bayes	0.6919	0.8335	0.5242	0.9155	0.6667	0.4742
Support Vector Machine	0.8578	0.8913	0.8361	0.7183	0.7727	0.6742

Observations

ML Model Name	Observation about model performance
Logistic Regression	Best overall accuracy, AUC, F1, and MCC. Scaling helps it combine the many correlated descriptors into a stable linear boundary.
Decision Tree	Lowest AUC among the tree-based and distance-based alternatives; its single, bounded-depth tree cannot represent all descriptor interactions as effectively.
KNN	Balanced performance, but weaker than the linear and kernel models. Distance-based decisions are sensitive to the high-dimensional descriptor space even after scaling.
Naive Bayes	Highest recall, so it identifies most RB compounds, but its low precision and accuracy indicate many false positives. The conditional-independence assumption is restrictive for molecular descriptors.
Support Vector Machine	Very strong AUC and precision, close to Logistic Regression. Its slightly lower recall leads to a marginally lower F1 and accuracy on this split.
Overall winner	**Logistic Regression, because it leads on the principal aggregate measures while remaining interpretable and efficient. SVM is a close second.**

7. Streamlit application features

1. A CSV uploader for labelled test data.
2. A classifier dropdown for all five saved models.
3. A Run evaluation button that evaluates the uploaded file only after explicit submission.
4. Accuracy, AUC, precision, recall, F1, and MCC metric cards.
5. A confusion matrix and prediction-summary chart.

6. Validation for missing columns and unsupported target labels.

Uploaded data must contain all 41 `descriptor_01` ... `descriptor_41` columns and a `target` column containing `NRB` or `RB`. The application loads the saved model and evaluates the uploaded rows without retraining.